

EUGENE L. BUCHANAN

Staff AI Engineer — Multi-Agent Frameworks & Agentic Infrastructure

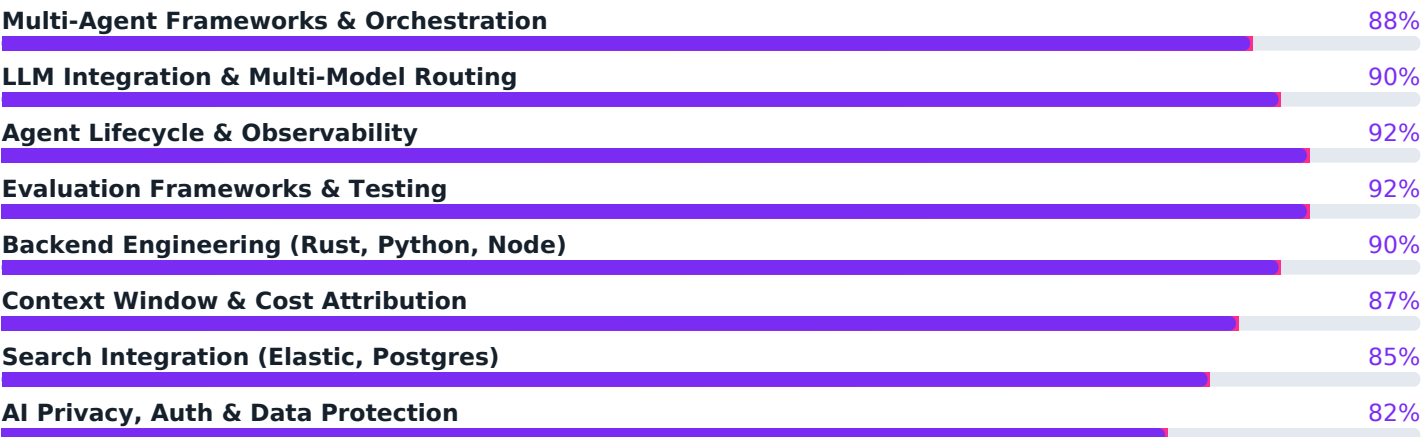


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Professional Summary

Staff AI engineer with 27 years software/systems experience focused on agentic AI infrastructure and multi-agent orchestration. Three production PRs to ZeroClaw (32.2k★ Rust gateway): merged context-window meter for 9 LLM providers, open PR for live provider identity on usage events (cost attribution), agent lifecycle observer via JSON-RPC. Built test automation, integration platforms, CI/CD pipelines on AWS/Postgres/React. Passionate about automatic model routing and platforms for deploying intelligent agents at scale.

Technical Expertise



Technology Stack

LangGraph Multi-Agent Coordination Agent Lifecycle JSON-RPC Bounded I/O Fire-and-Forget

Unix Domain Sockets

Context Window Routing Multi-Model Cost Attribution MCP (Model Context Protocol) herdr-mcp Server

External Tool Access Prompt Engineering OpenAI Anthropic Cohere Gemini

Rust Python Node.js PostgreSQL AWS (ECS, S3) REST GraphQL Docker Git

Cypress Selenium JUnit Postman Elasticsearch Regression Suites Health Dashboards Alerting

SAML OAuth OIDC Permission Profiles Principal Isolation Deny-by-Default

Experience

Staff AI Engineer / Agentic Infrastructure Contributor (Open-Source)

2026 – Present

ZeroClaw Labs (github.com/zeroclaw-labs/zeroclaw) · Rust · 32.2k stars

- PR #7946 (MERGED) — feat(runtime): model context window ctx bar across TUI/gateway/CLI (+1,109 / -22, 25 files). context_window as single source of truth for 9 providers; auto-populated config, doctor command, gateway API, live ContextBar. Eliminated drift between surfaces — directly relevant to Brain2's automatic model routing.
- PR #8966 (OPEN) — feat(agent): carry live provider identity on usage events (+2,742 / -141, 20 files). Fixed frozen 32k meter on 1M-token models; separated trim budget from model capacity; unconditional Usage events with serving-provider identity; per-provider usage breakdown on done frame — cost-attribution infra for multi-LLM orchestration.
- PR #8337 (OPEN) — feat(observability): herdr agent reporting integration (+1,712 / -17, 13 files). Observer reports agent lifecycle (idle/working/blocked/released) via JSON-RPC over UDS. Zero-config env detection, approval-gate blocked state, crash recovery via pane.release_agent. Multi-agent coordination infrastructure; maps to Super Agents' human-agent collaboration.

Senior Software Engineer / QA Lead / Integrations Product Owner

Dec 2020 – Aug 2025

Knowledgecity LLC · Remote · AWS / Postgres / React

- Product Owner, Integrations: shipped SAP, Oracle, Workday, Coursera, UKG, and Zoom integrations via SAML, OAuth, SFTP, and custom REST APIs — orchestrating multiple services with different auth models, data formats, and error semantics, the same coordination challenge as multi-agent systems.
- Built and maintained CI/CD quality pipelines with Cypress, Selenium, Postman, JUnit, Elastic, S3 — an evaluation framework for testing complex deployment scenarios across integration, backend, API, database, and frontend surfaces.
- Applied AI prompt engineering and chatbot support in production, integrating LLM-powered tooling into customer-facing assistance and internal automation workflows.
- Designed and implemented evaluation frameworks for regression suites, health dashboards, and alerting that measured system-level dynamics across multi-environment deployments.
- Mentored and trained test automation team; conducted code reviews and elevated engineering standards.

Senior Software Engineer II / Presales Engineering

1999 – 2001

RealNetworks · Seattle, WA · Streaming Media Platform

- QA on advanced research team focused on consumer appliances, mobile platforms, stream servers, and cellular networks — evaluating distributed system behavior in complex, multi-component scenarios.
- Global presales engineering: translating product capabilities into customer solutions across diverse technical environments. New-hire training and cross-functional collaboration.

Senior Software Test Engineer IV

1997 – 1999

Microsoft · Redmond, WA · Windows Media Server

- White-box testing/automation for Windows Media Server, Windows 98/NT — evaluation frameworks for complex system testing.
- Deep debugging across server infrastructure, network protocols, media codecs.

Education

- Associate Degree, Victor Valley College
- California Notary Commission
- Toastmasters International

Key Strengths

- **Multi-agent orchestration.** Shipped agent lifecycle observer (PR #8337) + MCP server in 32k-star Rust gateway — not a demo.
- **Multi-model routing & cost.** Merged context-window meter for 9 LLMs (PR #7946); open PR for live provider identity on usage events (PR #8966).
- **Evaluation frameworks.** 27 years test automation, regression suites, quality pipelines for complex distributed systems.
- **LLM integration.** Prompt engineering + chatbot tooling in production; multi-provider orchestration daily.

■ **Cross-functional collaboration.** Product Owner, QA Lead, presales engineer — comfortable with PMs, designers, engineers.